

1. SELENIUM TESTING

What is Selenium?

Definition: Web application testing framework for automated testing **Purpose:** Test web pages, forms, buttons, validation

Code Structure (5 Steps)

1. DECLARATION → Initialize browser driver & variables
2. BROWSER LAUNCH → Open target URL
3. FIELD INPUT → Enter values into form fields
4. SUBMIT ACTION → Click submit button
5. VERIFICATION → Check if submission was successful

URL Validation Logic

After Submit	Meaning
Same URL + Success message	Form submitted, page refreshed
Different URL (e.g., "Thank you")	Redirected = Success
Same URL + Error message	Validation failed

What to Test: Form fields (name, email, password, etc.), Submit button click, URL change after submission, Validation/error messages

2. SCRUM AND AGILE

What is Agility?

Definition: Capability to efficiently adapt to an ever-changing environment.

Three Drivers of Agility

Driver	Meaning
Flow	Steady, sustainable rate of work processing
Learning	Learn from past experiences, mistakes, knowledge discovery
Collaboration	Ways people work together for a single goal

Adaptive vs Predictive Approach

Aspect	Predictive (Waterfall)	Adaptive (Agile)
Focus	Planning future in detail	Flexibility to respond
Assumption	Stable, predictable environment	Fast, unpredictable changes
Phases	Sequential steps	Integrated, iterated
Output	Complete at end	Incremental deliverables

Agile Manifesto - 4 Core Values ☆ MEMORIZE

We Value	Over
Individuals and interactions	Processes and tools
Working software	Comprehensive documentation
Customer collaboration	Contract negotiation
Responding to change	Following a plan

Agile Manifesto - 12 Principles ☆

1. **Early & continuous delivery** of valuable software
2. **Welcome changing requirements**, even late in development
3. **Deliver working software frequently** (weeks, not months)
4. **Daily cooperation** between business and developers
5. Projects built around **motivated individuals** who are trusted
6. **Face-to-face conversation** is best communication
7. **Working software** = primary measure of progress
8. **Sustainable development** - maintain constant pace
9. **Continuous attention** to technical excellence
10. **Simplicity** - maximize work NOT done

11. Best work emerges from **self-organizing teams**
12. Team **reflects regularly** on how to improve

SCRUM FRAMEWORK ☆☆☆

Three Scrum Roles

Role	Responsibility
Product Owner	Represents stakeholders, manages product backlog, prioritizes features
Scrum Master	Manages Scrum process, removes impediments, protects team
Development Team	~7 people who develop the software (self-organizing)

Four Scrum Events (Ceremonies)

Event	Purpose	When	Duration
Sprint Planning	Plan work for upcoming sprint	Start of sprint	2-4 hours
Daily Scrum	Sync team, identify blockers	Daily	15 minutes
Sprint Review	Demo completed work to stakeholders	End of sprint	1-2 hours
Sprint Retrospective	Reflect on process, plan improvements	End of sprint	1-2 hours

Three Scrum Artifacts

Artifact	Description
Product Backlog	Ordered list of ALL requirements (user stories with story points)
Sprint Backlog	Items selected for CURRENT sprint
Increment	Sum of all completed items at sprint end

Sprint Workflow: Product Backlog → Sprint Planning → Sprint Backlog → Daily Scrum → Development → Sprint Review → Sprint Retrospective → REPEAT

Product Backlog: It is a **main list of everything** that needs to be built in a product (features, fixes, improvements).

Example:

- There are 10 tasks (user stories).
- Each task has story points to show how difficult it is.
- The team can complete 10 points in one sprint.
- One sprint lasts 1 week.

Why important? It helps the team decide what to build first.

Sprint Backlog: It is the list of tasks the team chooses to finish **in the current sprint**.

Example:

- For Sprint 1, the team picked **3 tasks** that match their capacity.

Why important? Keeps the team focused and prevents taking too much work.

Daily Progress (Simple View)

- **Day 1:** Work started, nothing finished.
- **Day 2:** One task completed.
- **Day 3:** Two tasks completed, one remaining.
- **Day 4:** Last task almost done.
- **Day 5:** All tasks finished ☒

👉 This means the sprint was successful.

Important Terms

- ✓ **Story Points:** Show how hard a task is (not how long it takes).
- ✓ **Velocity:** How much work a team finishes in one sprint.
- ✓ **Sprint:** A short time (1–4 weeks) to complete tasks.

✓ **In Progress:** Currently being worked on.

✓ **Done:** Fully completed.

SCRUM VS KANBAN

Aspect	Scrum	Kanban
Scheduling	Fixed sprints with deadlines	Continuous flow
Roles	Specific duties (PO, SM, Team)	Not strictly determined
Changes	No changes mid-sprint	Encourages modifications
KPIs	Sprint velocity	Time to completion
Best For	Fixed deliverables	Flexible teams

SCENARIO QUESTIONS - How to Answer

Question Type: "In a project, X happened. What should PM do?"

Answer Approach: Identify which Scrum activity/event is relevant, Explain what **SHOULD** have been done, State the correct Scrum process

Example Scenario: "Team is confused about what to work on. What went wrong?"

Answer: Sprint Planning was not done properly. Product Owner should have clearly defined Sprint Backlog with prioritized items.

3. COCOMO ESTIMATION

What is COCOMO? COConstructive COst MOdel - **Software cost estimation model**

4. PROCUREMENT CONTRACTS

What is Procurement?

Obtaining goods/services from outside vendors (Supplier/Contractor/Seller).

Four Procurement Processes

Process	Description
Plan Procurement	Identify what to outsource
Conduct Procurement	RFI → RFP → Vendor Selection
Control Procurement	Monitor performance, three-way matching
Close Procurement	Final payment, contract closure

THREE MAIN CONTRACT TYPES

1. FIXED-PRICE CONTRACT

Key Characteristics: Price fixed at contract signing, **Seller bears most risk**, Best for well-defined scope, No price change unless scope changes

Subdivisions:

Type	Description	Risk
Firm Fixed Price (FFP)	Fixed amount, no adjustments	All on Seller
Fixed Price Incentive Fee (FPIF)	Fixed price + bonus for good performance	Mostly Seller
FP-EPA	Multi-year with inflation adjustment	Shared

Use When: Scope is CLEAR, Requirements are DEFINED, Want to CONTROL COST

Examples: Purchasing apartment, Off-the-shelf software, Holiday package (all-inclusive), Road construction

2. COST REIMBURSABLE CONTRACT

Key Characteristics: Buyer reimburses ALL costs + fee, **Buyer bears most risk**, Best for uncertain/evolving scope, Watch for scope creep!

Subdivisions:

Type	Description	Fee
Cost Plus Fixed Fee (CPFF)	All costs + fixed fee	Fixed regardless of performance
Cost Plus Incentive Fee (CPIF)	All costs + incentive	Based on objective metrics
Cost Plus Award Fee (CPAF)	All costs + award	Based on subjective satisfaction

Use When: Scope is UNCLEAR, Requirements will EVOLVE, R&D or creative projects, Need specialized vendor skills

Examples: Research & Development, Event decoration, Media campaigns

3. TIME AND MATERIALS (T&M) CONTRACT

Key Characteristics:

- Pay for time spent + materials used + profit
- **Risk shared** between buyer and seller
- Hybrid of fixed-price and cost-reimbursable

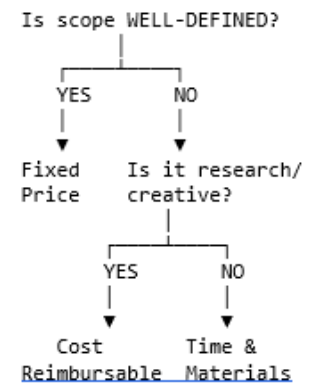
Components:

- **Time:** Hourly/daily rate
- **Materials:** Physical inputs

Use When: Moderate certainty, Scope partially defined, Support and maintenance work

Examples: Post-launch support, Staff augmentation, Consulting services

CONTRACT SELECTION DECISION FLOW



SCENARIO: Development + Support Contract

Situation: Company needs both software development AND ongoing support.

Analysis:

Aspect	Development	Support
Scope	Clear deliverables	Unknown workload
Duration	Fixed timeline	Ongoing
Effort	Predictable	Varies (10 hrs to 1000 hrs/month)
Contract	**Fixed Price**	**Time & Materials**

Result: TWO different contracts needed!

REAL-TIME PROCUREMENT SCENARIO (Mobile Banking App)

Phase 1: Plan Procurement

- Mobile app development (Android & iOS)
- Cybersecurity testing
- Cloud hosting
- UI/UX design

Phase 2: Conduct Procurement

- RFI → Understand vendor capabilities
- RFP → Get technical + financial proposals
- Select Vendor ABC Technologies

Phase 3: Contracts Used

Work	Contract Type	Reason
Core App Development	FFP (\$200,000)	Scope is clear
Early Delivery Bonus	FPIF (\$15,000)	Incentive
Cybersecurity Testing	CPFF	Scope uncertain (threats evolve)
Post-Launch Support	T&M (\$50-70/hr)	Effort varies
Cloud Services	PO (\$10,000/month)	Commodity purchase

QUICK REFERENCE TABLE

Scenario	Contract Type
Clear scope, fixed requirements	FFP
Want early delivery incentive	FPIF

Multi-year with inflation	FP-EPA
R&D, uncertain scope	CPFF/CPIF
Creative work, client satisfaction matters	CPAF
Support/maintenance	T&M
Buying commodities	Purchase Order

5. PM RESPONSIBILITIES

What is Project Management?

Application of **knowledge, skills, tools, and techniques** to meet project requirements.

8 Core Management Activities

Activity	Description
Planning	Deciding what is to be done
Organizing	Making arrangements
Staffing	Selecting right people
Directing	Giving instructions
Monitoring	Checking on progress
Controlling	Taking action to remedy hold-ups
Innovating	Coming up with new solutions
Representing	Liaising with users

5 Process Groups

Process	Description
Initiating	Project gets selected, sponsored, launched
Planning	Determine HOW work will be accomplished
Executing	Team does the work
Monitoring & Controlling	Track, review, regulate progress
Closing	Tie up loose ends, close project

10 Knowledge Areas

#	Knowledge Area	Key Focus
1	**Integration**	Coordinate all processes
2	**Scope**	Define boundaries, what's included/excluded
3	**Time**	Scheduling, deadlines
4	**Cost**	Budgeting, cost control
5	**Quality**	Standards, QA/QC
6	**Human Resource**	Team management, motivation
7	**Communications**	Information distribution
8	**Risk**	Identify, analyze, respond to risks
9	**Procurement**	Acquire external goods/services
10	**Stakeholder**	Manage expectations

PM STRATEGY PARAGRAPH ☆ (Professor's Template)

****Building a software project naturally involves uncertainty, evolving requirements, and need for strong communication. As PM, my role revolves around applying core management responsibilities:****

1. Stakeholder Management: Communicate clearly about realistic deliverables. Set expectations early that software may need iterative refinement.

2. Team Coordination: Use core activities - planning, organizing, directing, monitoring, controlling. Break work into manageable tasks, clarify responsibilities, hold daily check-ins.

3. Change Management: Planning is iterative. Maintain prioritized requirements list, evaluate impact of changes on time/cost, update plan accordingly. No change accepted without checking feasibility.

4. Communication: Document all project information (requirements, decisions, progress) transparently. Establish single source of truth.

Summary: By communicating clearly, planning iteratively, coordinating team, managing change, and monitoring progress—PM guides project through shifting requirements while keeping stakeholders confident.

6. QUALITY MANAGEMENT

Three Quality Management Processes

Process	Description
Quality Planning	Define quality standards, metrics, create QM plan
Quality Assurance (QA)	Process audits, continuous improvement, PREVENTION
Quality Control (QC)	Inspection, testing, defect identification, DETECTION

QA vs QC

Aspect	Quality Assurance	Quality Control
Focus	Process	Product
When	During development	After development
Purpose	Prevent defects	Detect defects
Approach	Proactive	Reactive

Key Quality Concepts

Term	Definition
Quality	Degree to which product meets requirements
Grade	Category assigned to deliverables (High/Low)
Precision	Consistency of measurements
Accuracy	Correctness of measurements

⚠ **Note:** Professor will provide 3-4 slides before exam

7. CONFIGURATION MANAGEMENT

What is Configuration Management?

Managing and tracking changes to project artifacts (code, documents, deliverables).

Key Concepts

Concept	Description
Version Control	Track all changes to files
Configuration Item	Element under configuration control
Baseline	Approved version at a point in time
Change Control	Process to manage changes to baseline

Configuration Management Activities

1. **Identification** - Identify items to control
2. **Control** - Manage changes
3. **Status Accounting** - Track status of items
4. **Audit** - Verify integrity of items

⚠ **Note:** Professor will provide overview slides

9. COMMUNICATION MANAGEMENT

Communication Channels Formula

Number of Channels = $n(n-1) / 2$ Where n = number of stakeholders/team members

Examples:

Team Size (n)	Channels	Calculation
5	10	$5 \times 4 / 2 = 10$
10	45	$10 \times 9 / 2 = 45$
15	105	$15 \times 14 / 2 = 105$
20	190	$20 \times 19 / 2 = 190$

Effect of Adding Team Members

Problem: Team has 10 members. 3 more join. How many NEW channels?

Original channels: $10(10-1)/2 = 45$

New team size: $10 + 3 = 13$

New channels: $13(13-1)/2 = 78$

Additional channels: $78 - 45 = 33$ new channels

Three Communication Methods

Method	Description	Examples
Interactive	Real-time, multi-directional	Meetings, calls, video conferences
Push	One-way, sent to specific receivers	Emails, memos, reports
Pull	Receiver accesses when needed	Intranet, shared drives, wikis

STRATEGIES FOR REDUCING COMMUNICATION COMPLEXITY ☆

****Professor's Paragraph - Use as Template:****

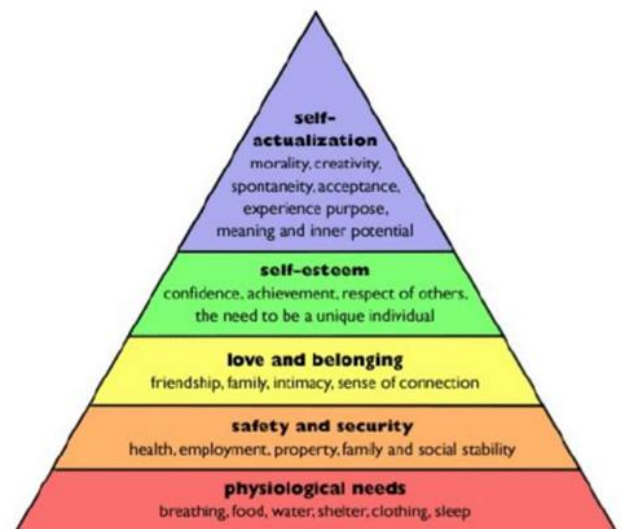
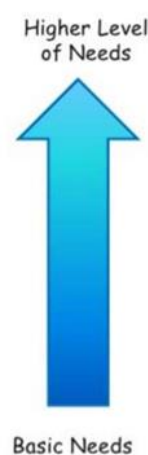
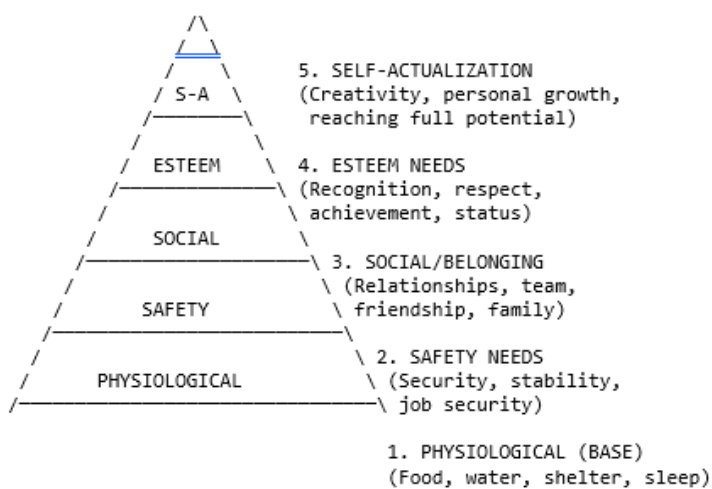
Too many communication channels can lead to confusion. PM can simplify by:

1. ****Create sub-teams**** - Let each group communicate through a single representative
2. ****Set up clear communication plan**** - Define who talks to whom, how often, which tools
3. ****Single source of truth**** - Keep all information in one project management tool
4. ****Reduce meeting overload**** - Combine similar meetings, remove low-value ones
5. ****Limit direct stakeholder access**** - Route through PM or team leads only

These actions make communication cleaner, simpler, and easier for everyone.

10. MOTIVATION THEORIES

MASLOW'S HIERARCHY OF NEEDS ☆ ☆



Key Points:

- People move UP as lower needs are satisfied
- Cannot focus on higher needs if lower needs are unmet
- PM must identify which level employee is at

Level Examples:

Level	Examples in Workplace
Physiological	Salary for food/shelter, breaks, comfortable workspace
Safety	Job security, safe work environment, health insurance
Social	Team activities, good relationships, belonging to team
Esteem	Recognition, promotions, respect from colleagues
Self-Actualization	Challenging work, creativity, personal development

HERZBERG'S TWO-FACTOR THEORY ☆☆

Factor Type	Effect	Examples
Hygiene Factors (Dissatisfiers)	Cause dissatisfaction if MISSING, but DON'T motivate if present	Salary, policies, working conditions, job security, relationships, status
Motivation Factors (Satisfiers)	TRULY motivate and increase satisfaction	Achievement, recognition, the work itself, responsibility, growth, advancement

Key Insight ☆

Hygiene factors met ≠ Motivation. A person can have good salary, house, car, family BUT still be unmotivated! True motivation comes from MOTIVATION FACTORS.

SCENARIO EXAMPLE 1: Maslow

Question: "A project manager has a good house, good car, and happy family. But he is not happy at work. Explain why using Maslow's theory."

Answer:

Using Maslow's Hierarchy of Needs:

1. **Physiological needs** - MET (has salary for food, shelter)
2. **Safety needs** - MET (job security, stable life)
3. **Social needs** - MET (happy family, relationships)
4. **Esteem needs** - MAY NOT BE MET (needs recognition, achievement, respect)
5. **Self-actualization** - NOT MET (needs personal growth, challenging work)

The PM is unhappy because lower-level needs are satisfied, but higher-level needs (esteem, self-actualization) are NOT met. He needs: Recognition for his work, Challenging projects, Opportunities for growth, Respect from peers

SCENARIO EXAMPLE 2: Herzberg

Question: "An employee has all material comforts - good salary, nice office, job security. How would you motivate this person?"

Answer: Using Herzberg's Two-Factor Theory:

Hygiene factors present: Good salary, Nice office (working conditions), Job security

These only PREVENT dissatisfaction but don't MOTIVATE.

To truly motivate, provide Motivation Factors:

1. **Give challenging work** - interesting, meaningful tasks
2. **Provide recognition** - acknowledge achievements publicly
3. **Offer growth opportunities** - training, new skills, career advancement
4. **Increase responsibility** - delegate important decisions
5. **Enable achievement** - set goals that lead to accomplishment

QUICK COMPARISON

Aspect	Maslow	Herzberg
Focus	Human needs hierarchy	Job satisfaction factors
Levels	5 levels (pyramid)	2 categories
Key Idea	Satisfy lower needs first	Hygiene ≠ Motivation
Application	Identify which level person is at	Identify if missing hygiene or motivation factors

12. STAKEHOLDER REGISTER

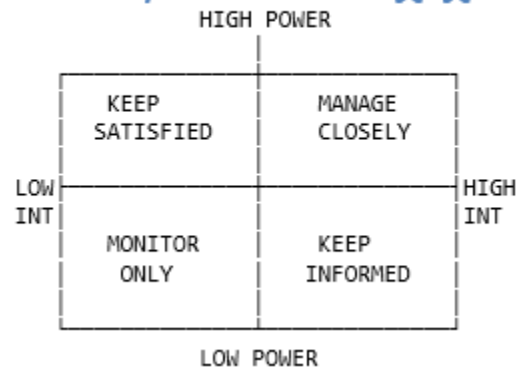
What is a Stakeholder?

Individuals, groups, or organizations who may affect, be affected by, or perceive themselves affected by the project.

Stakeholder Register Components

Column	Description
Name/Role	Stakeholder identification
Job Title	Position in organization
Contact	Phone, email
Category	Internal/External
Expectation	What they want from project
Communication Req.	How/when to update them
Power/Interest	Grid position
Engagement Level	Current & Desired

POWER/INTEREST GRID ☆☆



Four Quadrants

Quadrant	Power	Interest	Strategy	Examples
Manage Closely	High	High	Work closely, full engagement	Sponsor, PM, Key client
Keep Satisfied	High	Low	Keep happy, don't bore with details	Senior executives, Regulators
Keep Informed	Low	High	Regular updates, can help with details	End-users, Team members
Monitor	Low	Low	Minimal effort, newsletters	General public

STAKEHOLDER REGISTER EXAMPLE

Name	Job Title	Contact	Category	Expectation	Communication	Power/Interest	Engagement
Qasim	Project Lead	0333-2244768	Internal	On time, within budget	Email & Tel	Keep Satisfied	Leading
Ali	Sponsor	ali@company.com	Internal	ROI, success	Weekly report	Manage Closely	Supportive
Users	End Users	-	External	Easy to use	Monthly demo	Keep Informed	Neutral

LEVELS OF ENGAGEMENT

Level	Description
Unaware	Doesn't know project exists
Resistant	Doesn't want project to happen
Neutral	Fine either way
Supportive	Wants project to succeed
Leading	Actively helping project succeed

13. ORGANIZATIONAL STRUCTURE

Three Main Types

Structure	PM Authority	Func. Mgr Authority	Team Reporting
Functional	Very Low	Very High	To Func. Manager
Projectized	Very High	Very Low	To PM
Matrix	Varies	Varies	Dual reporting

MATRIX STRUCTURE VARIANTS ☆ ☆

Type	PM Authority	Func. Mgr Authority	Conflict Level
Weak Matrix	20-30%	70-80%	Medium
Balanced Matrix	50%	50%	**HIGHEST**
Strong Matrix	70-80%	20-30%	Medium

Why Balanced Matrix Has HIGHEST Conflicts?

- Equal power = both want control
- Team member has TWO bosses with equal authority
- Competing priorities
- Unclear decision-making authority

COMPLETE COMPARISON TABLE

Aspect	Functional	Weak Matrix	Balanced Matrix	Strong Matrix	Projectized
PM Authority	None/Little	Limited	Low-Moderate	Moderate-High	High-Total
Func. Mgr Role	Controls	Controls	Mixed	Limited	Little/None
PM Role	Part-time	Part-time	Full-time	Full-time	Full-time
Resource Avail.	Little	Limited	Low-Moderate	Moderate-High	High-Total
Budget Control	Func. Mgr	Func. Mgr	Mixed	PM	PM
PM Admin Staff	Part-time	Part-time	Part-time	Full-time	Full-time

SCENARIO QUESTION

Question: "In an organization, team members report to both project manager and department heads. Equal power distribution creates frequent conflicts. What structure is this?"

Answer: This is a **Balanced (Standard) Matrix** structure because: Dual reporting to both PM and Functional Manager. Equal (50-50) authority distribution. Highest conflict level due to equal power. Neither manager dominates decisions